Kernix 0.10.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

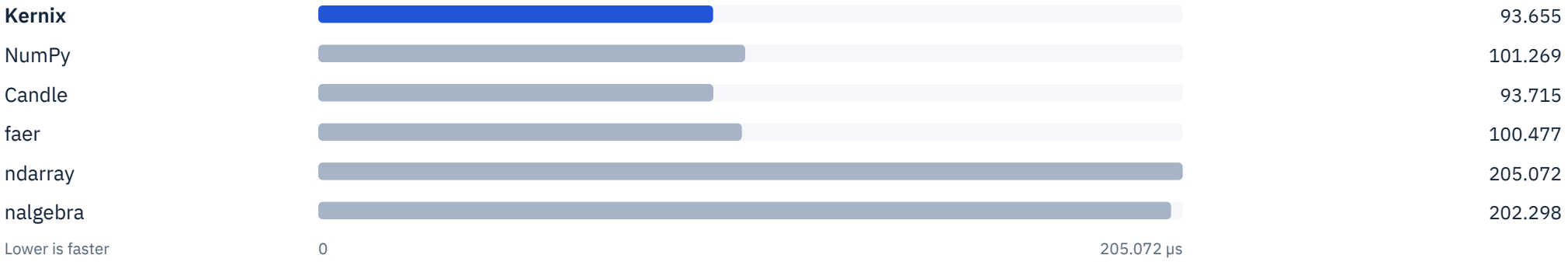
31 workloads

108 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

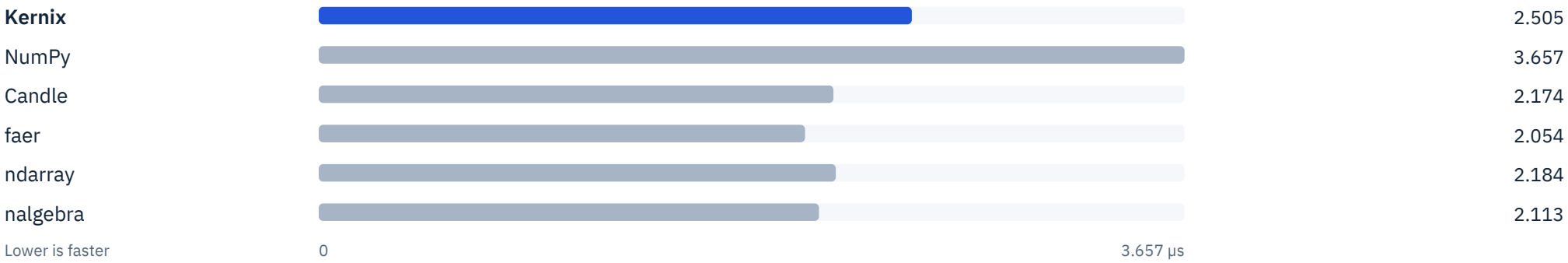
Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

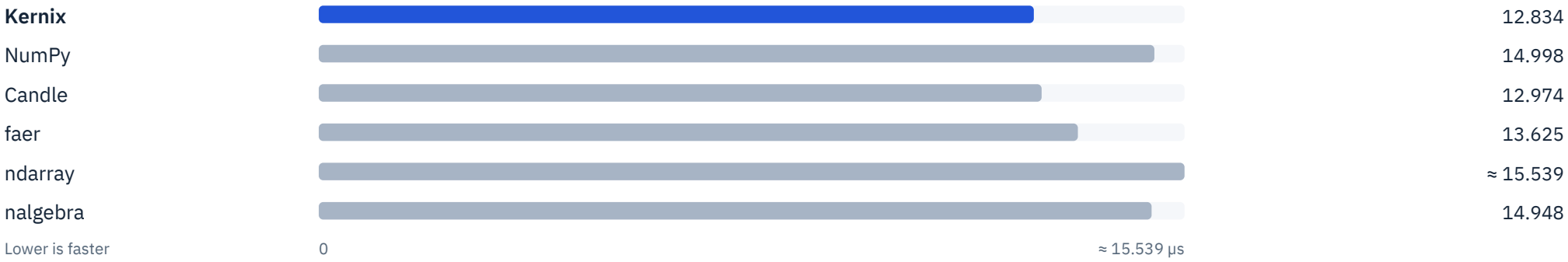
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



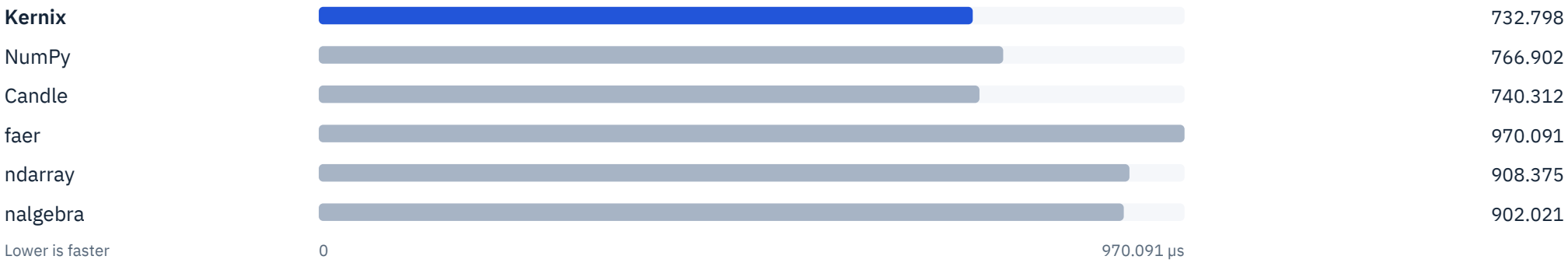
Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

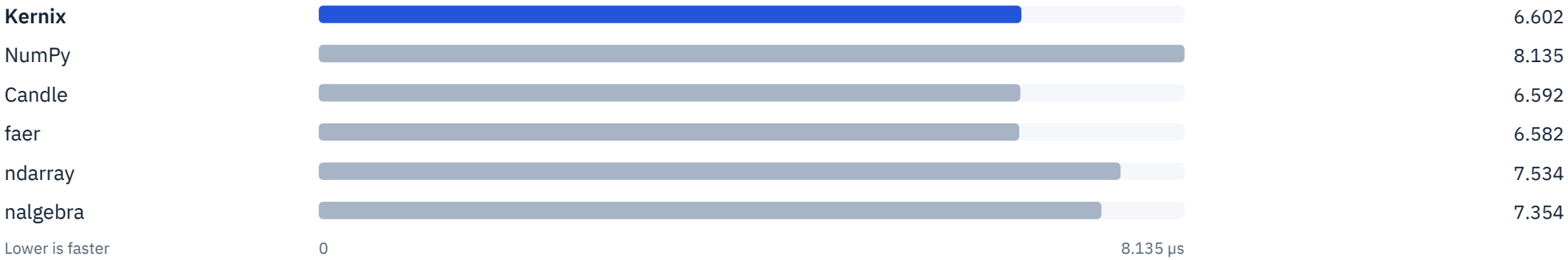
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



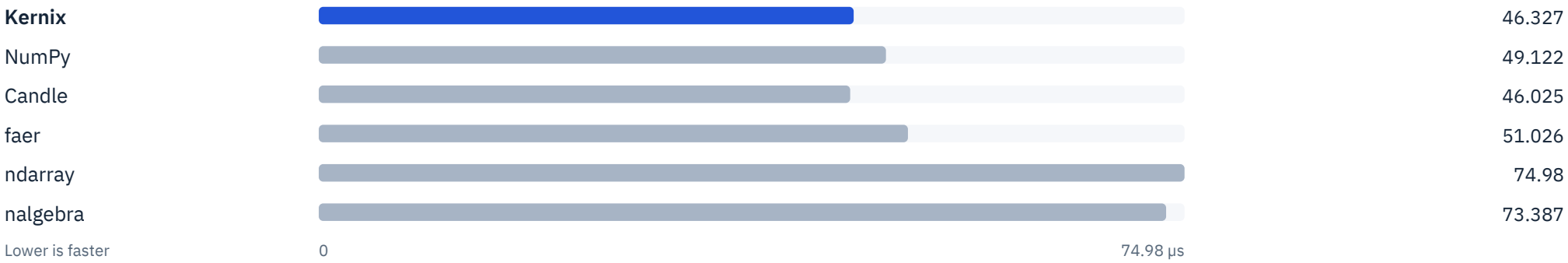
Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

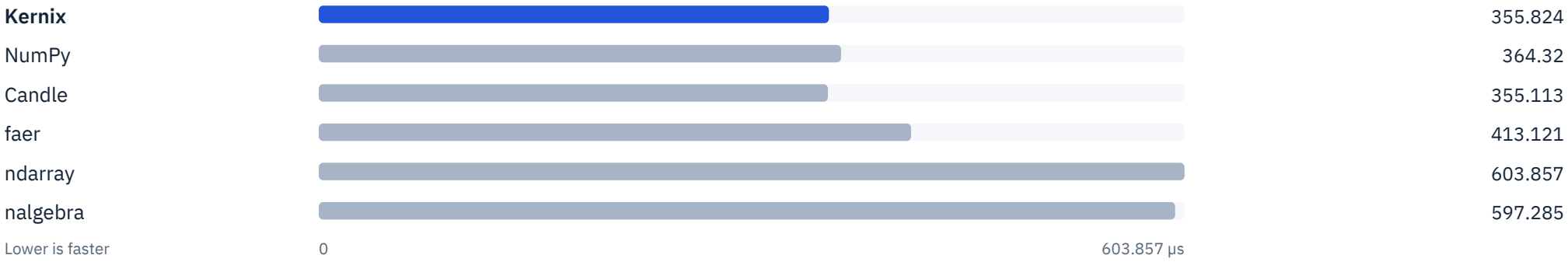
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

Per-call median



Kernix 0.10.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

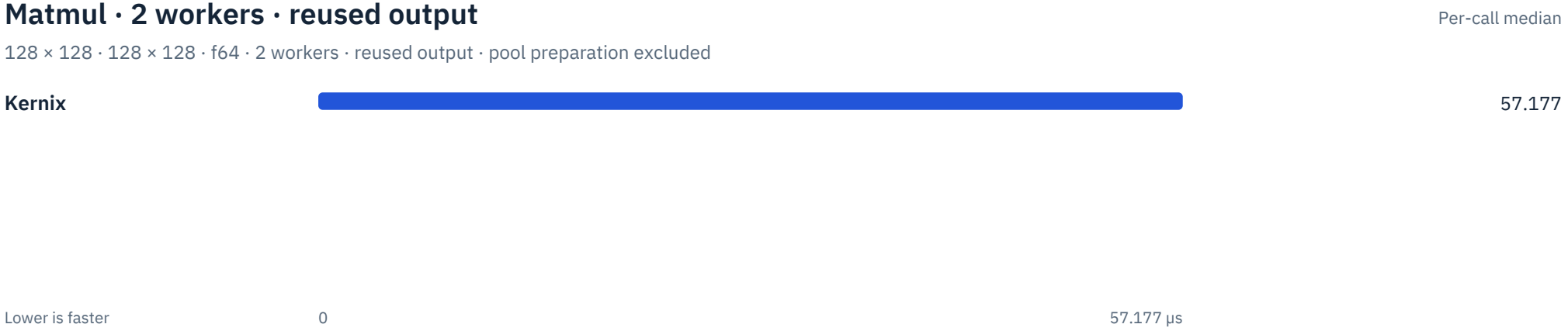
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.10.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



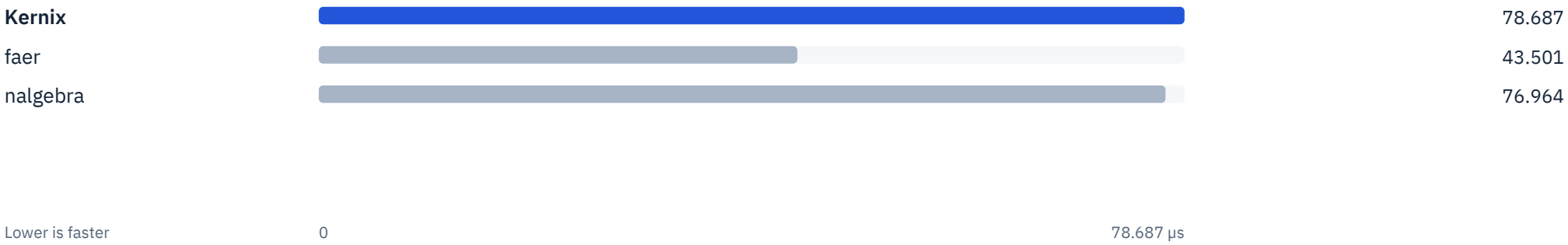
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

Per-call median

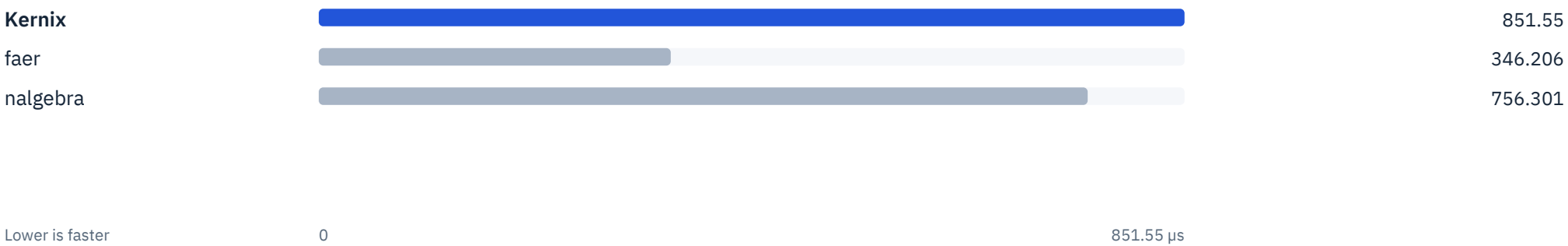
96 × 32 · f64 · 1 worker · allocated output



Column-pivoted QR

Per-call median

256 × 64 · f64 · 1 worker · allocated output



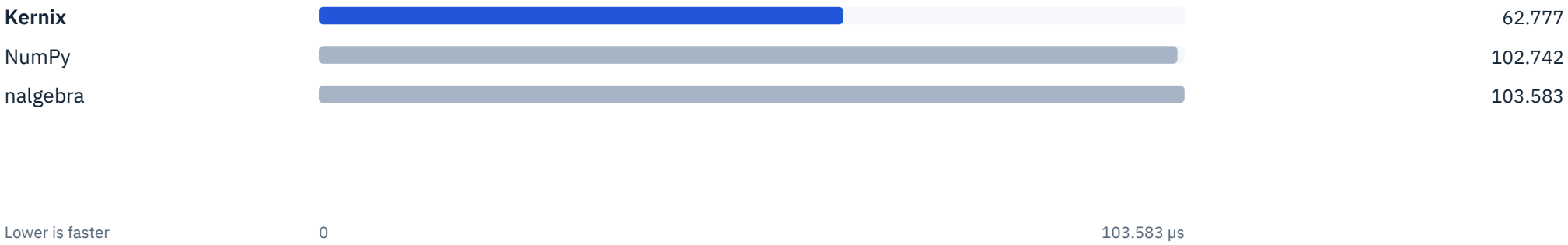
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

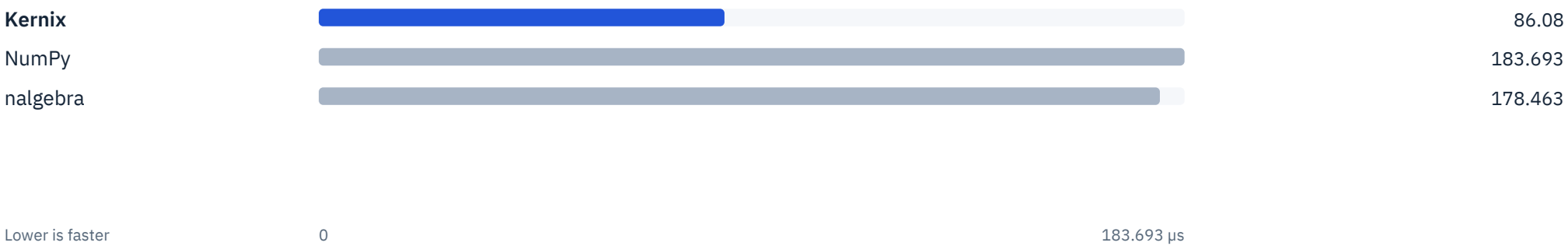
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

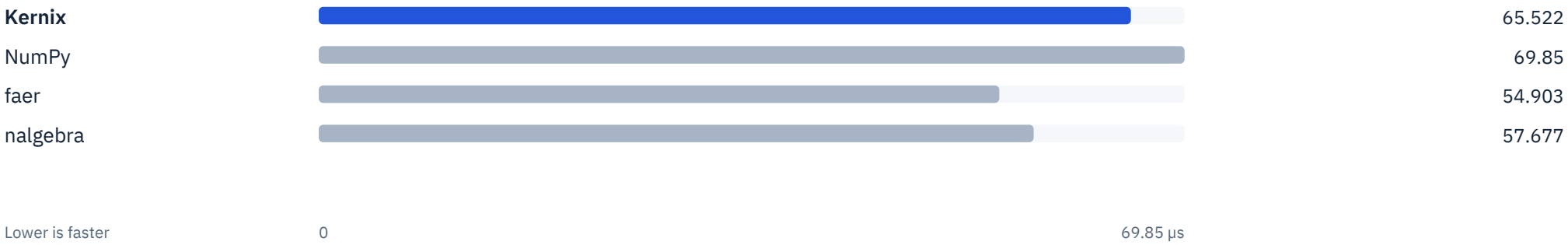


Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

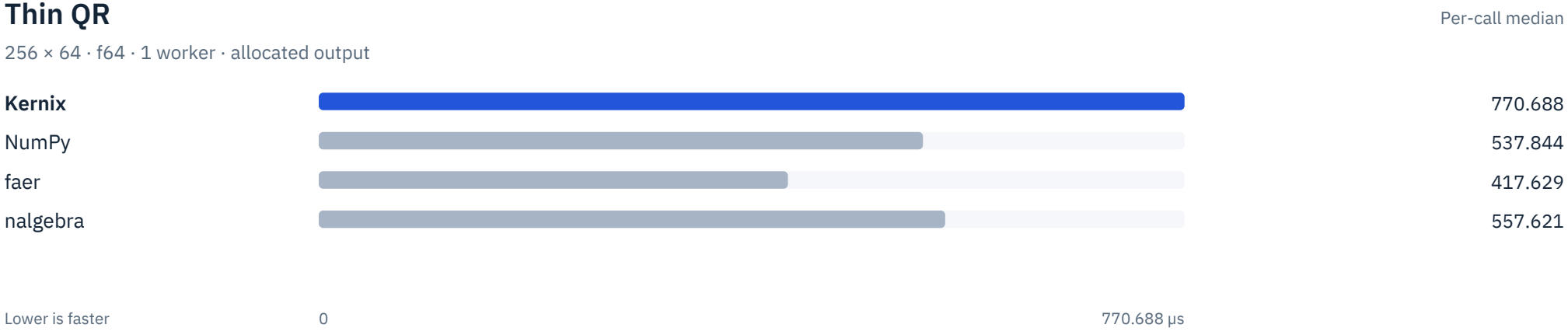
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



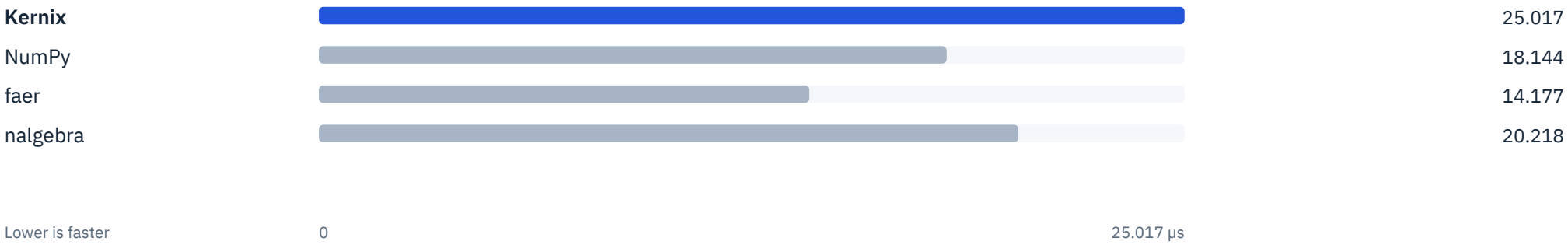
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

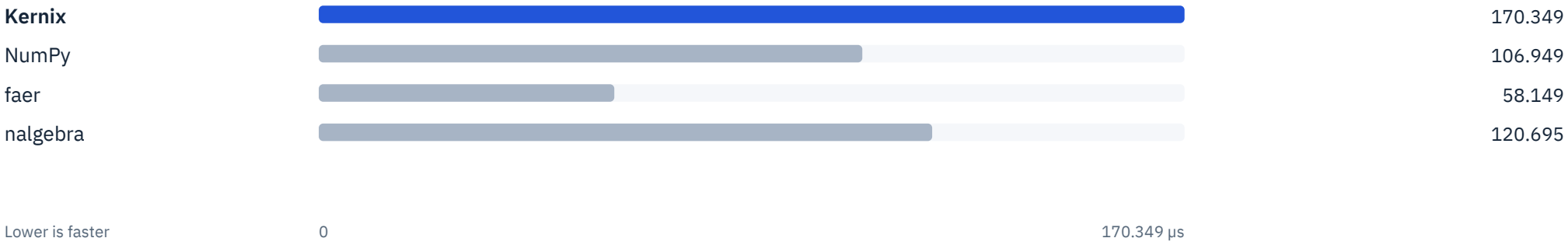
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



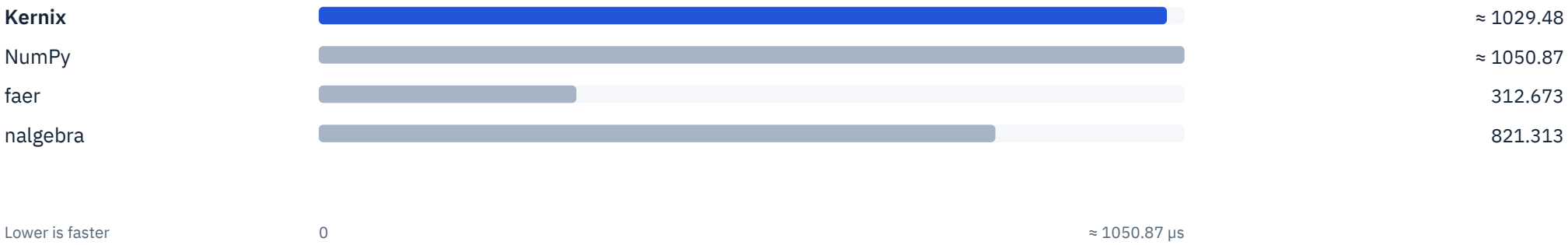
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



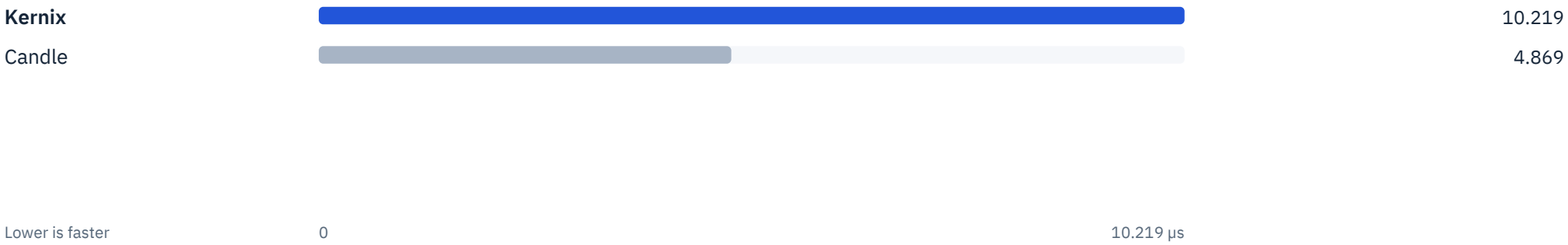
Kernix 0.10.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.10.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

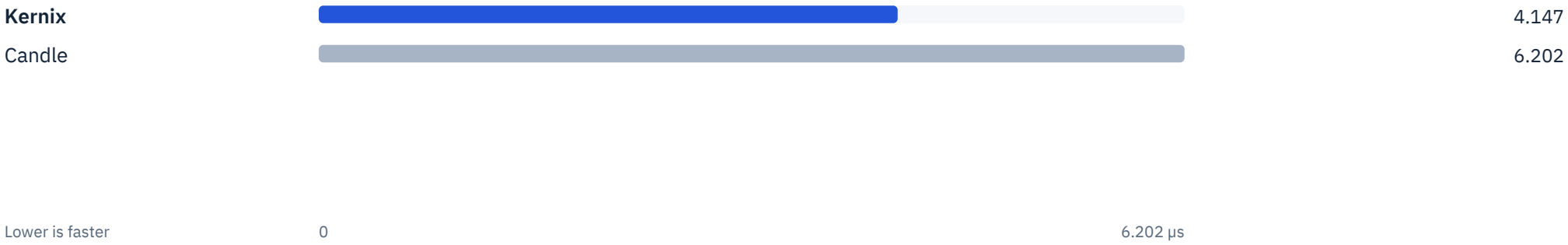
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

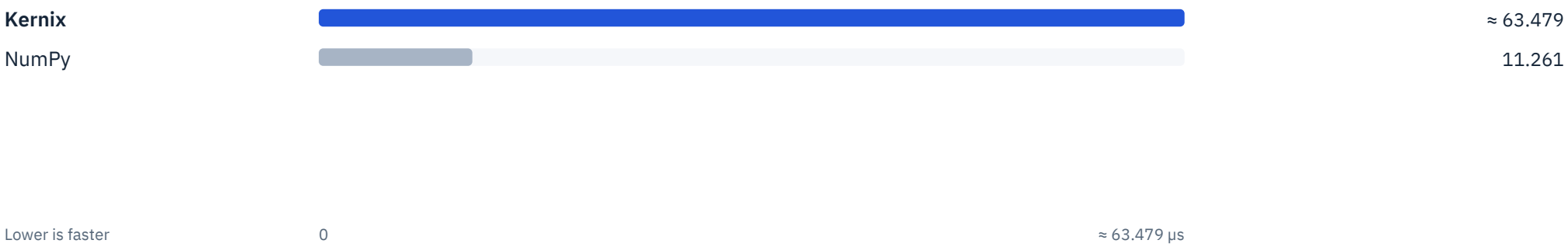


Kernix 0.10.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



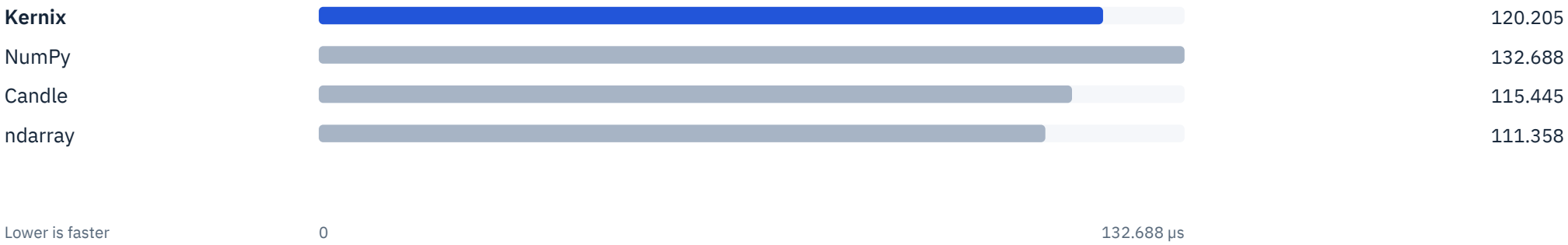
Kernix 0.10.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Per-call median

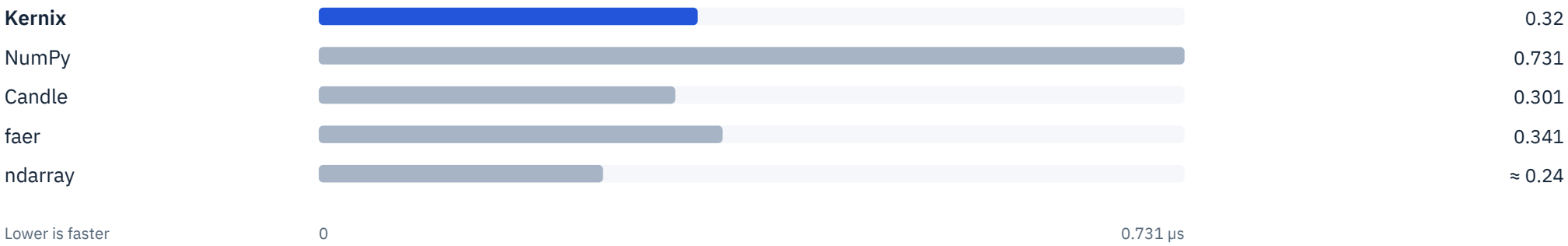


Kernix 0.10.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2054	2.054
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2174	2.174
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	3657	3.657
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2505	2.505
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2184	2.184
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2113	2.113
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	15539.000000000002	15.539000000000002
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	13625	13.625
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	12974	12.974
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12834	12.834

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	14998	14.998
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	14948	14.948
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	93655	93.655
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	100477	100.477
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	205072	205.072
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	202298	202.298
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	93715	93.715
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	101269	101.269
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	732798	732.798
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	766902	766.902

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	740312	740.312
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	970091	970.091
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	902021	902.021
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	908375	908.375
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7354	7.354
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	6592	6.592
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6602	6.602
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	6582	6.582
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7534	7.534
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	8135	8.135

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	51026	51.026
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	49122	49.122
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	46025	46.025
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	74980	74.98
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	46327	46.327
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	73387	73.387
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	413121	413.121
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	355824	355.824
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	355113	355.113
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	597285	597.285

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	603857	603.857
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	364320	364.32
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3276	3.276
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	4197	4.197
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3086	3.086
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	16601	16.601
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	57177	57.177
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	395459	395.459
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	31619.000000000004	31.619000000000004
Column-pivoted QR / Kernix 96 × 32 · f64	78687	78.687

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	43501	43.501
Column-pivoted QR / nalgebra 96 × 32 · f64	76964	76.964
Column-pivoted QR / Kernix 256 × 64 · f64	851550	851.55
Column-pivoted QR / faer 256 × 64 · f64	346206	346.206
Column-pivoted QR / nalgebra 256 × 64 · f64	756301	756.301
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	62777	62.777
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	103583	103.583
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	102742	102.742
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	178463	178.463
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	86080	86.08

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	183693	183.693
Thin QR / nalgebra 96 × 32 · f64	57677	57.677
Thin QR / NumPy 96 × 32 · f64	69850	69.85
Thin QR / Kernix 96 × 32 · f64	65522	65.522
Thin QR / faer 96 × 32 · f64	54903	54.903
Thin QR / faer 256 × 64 · f64	417629	417.629
Thin QR / Kernix 256 × 64 · f64	770688	770.688
Thin QR / NumPy 256 × 64 · f64	537844	537.844
Thin QR / nalgebra 256 × 64 · f64	557621	557.621
Cholesky factorization / NumPy 64 × 64 · f64	18144	18.144

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	25017	25.017
Cholesky factorization / faer 64 × 64 · f64	14177	14.177
Cholesky factorization / nalgebra 64 × 64 · f64	20218	20.218
Cholesky factorization / nalgebra 128 × 128 · f64	120695	120.695
Cholesky factorization / NumPy 128 × 128 · f64	106949	106.949
Cholesky factorization / faer 128 × 128 · f64	58149	58.149
Cholesky factorization / Kernix 128 × 128 · f64	170349	170.349
Cholesky factorization / nalgebra 256 × 256 · f64	821313	821.313
Cholesky factorization / faer 256 × 256 · f64	312673	312.673
Cholesky factorization / Kernix 256 × 256 · f64	1029482	1029.482

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 256 × 256 · f64	1050872	1050.872
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4869	4.869
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	10219	10.219
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7584	7.584
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	4107	4.107
Model inference / Candle batched normalized/sliced MLP · f64	4979	4.979
Model inference / Kernix batched normalized/sliced MLP · f64	3236	3.236
Model recording + forward / Candle batched normalized/sliced MLP · f64	6202	6.202
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4147	4.147
Model backward / Kernix prepared graph → fresh gradients · f64	5139	5.139

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Candle prepared graph → fresh gradients · f64	20879	20.879
Parameter update / Kernix transactional SGD, fixed gradients · f64	801	0.801
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63479.00000000001	63.47900000000001
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	11261	11.261
Row softmax / Candle 256 × 64 · f64	115445	115.445
Row softmax / ndarray 256 × 64 · f64	111358	111.358
Row softmax / Kernix 256 × 64 · f64	120205	120.205
Row softmax / NumPy 256 × 64 · f64	132688	132.688
Create zeros / Candle 64 × 64 · f32	301	0.301
Create zeros / faer 64 × 64 · f32	341	0.341

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	320	0.32
Create zeros / ndarray 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create zeros / NumPy 64 × 64 · f32	731	0.731
Create ones / Kernix 64 × 64 · f32	431	0.431
Create ones / Candle 64 × 64 · f32	461	0.461
Create ones / NumPy 64 × 64 · f32	2084	2.084
Create ones / ndarray 64 × 64 · f32	361	0.361
Create ones / faer 64 × 64 · f32	391	0.391